

Physics A (H156, H556)

Exploring - Fundamental particles ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

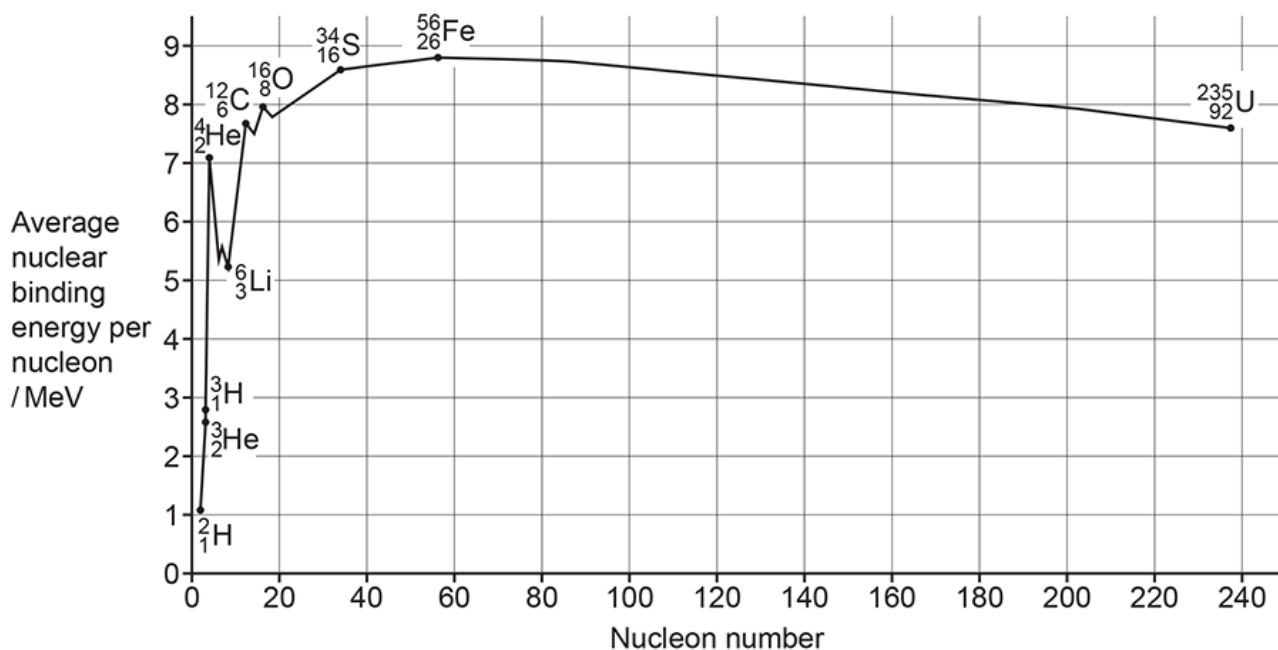
INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **47**.
- The total number of marks may take into account some 'either/or' question choices.

1(a) The diagram below shows the average nuclear binding energy per nucleon for a number of different isotopes.



In the Sun, deuterium (^2_1H) is produced from fusion of two hydrogen (^1_1H) nuclei, as shown below.

There is a particle missing



i. Determine the charge of the missing particle.

----- [1]

ii. The missing particle is a lepton. Name this lepton.

----- [1]

iii. In the fusion reaction above, determine the total number of up quarks at the **start** of the reaction.

----- [1]

- (b) Tritium (${}^3_1\text{H}$) is another isotope of hydrogen which is formed in stars. On the Earth, tritium is a radioactive element which decays by β emission.

Write down the equation for β - decay in terms of quarks.

[2]

- 2 A sub-atomic particle has a positive charge.

Which type of particle is it?

- A anti-proton
- B down quark
- C neutrino
- D positron

Your answer

[1]

- 3 A neutrino is a fundamental particle.

Which row of the table correctly describes a neutrino?

	Classification	Force felt
A	hadron	strong nuclear
B	hadron	weak nuclear
C	lepton	strong nuclear
D	lepton	weak nuclear

Your answer

[1]

- 4 Which one of these non-invasive medical scans does **not** expose the patient to ionising radiation?.

- A CAT
- B PET
- C Ultrasound
- D X-ray

Your answer

[1]

- 5 Which of these statements is/are true?
- 1 Antiprotons are hadrons so are subject to the strong nuclear and weak nuclear forces.
 - 2 Neutrons are subject to the weak nuclear force only.
 - 3 The weak nuclear force is the only force that causes a change of quark type.
- A 1, 2 and 3
 B Only 1 and 2
 C Only 1 and 3
 D Only 3

Your answer

[1]

6(a)

Deuterium is an isotope of hydrogen.

A nucleus of deuterium has a proton and a neutron.

Describe the nature and range of the **two** forces acting between these two hadrons.

[4]

(b)

*Here is some data for a nucleus of carbon-14 (${}^{14}_6\text{C}$) and a nucleus of uranium-235 (${}^{235}_{92}\text{U}$).

	Carbon-14 nucleus	Uranium-235 nucleus
Decay mode	Beta-minus decay	Alpha decay
Mass of nucleus / u	14.0	235.0
Radius of nucleus / 10^{-15}m	2.9	7.4

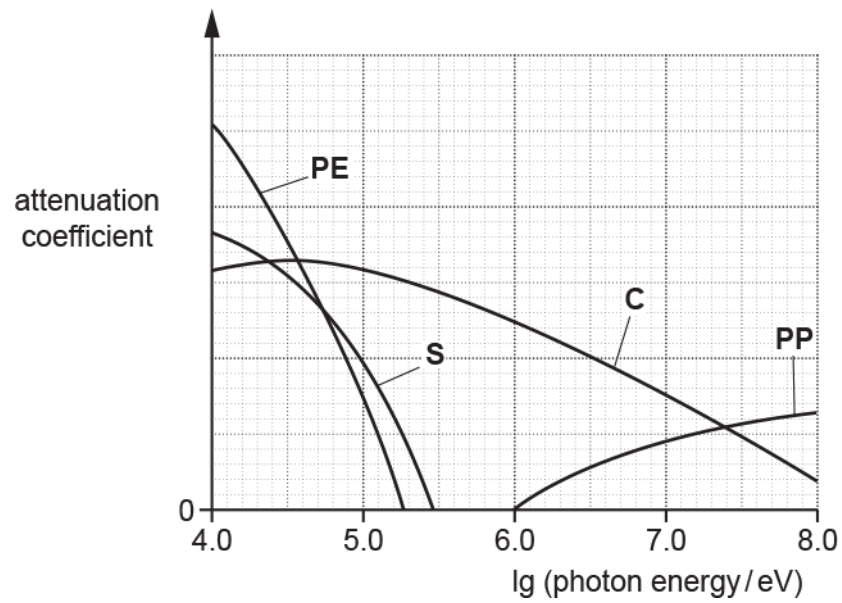
Use the data to:

- describe the composition of the nuclei before and after the decay in terms of hadrons and quarks
- show that both nuclei have the same density.

Blank lined paper for writing.

7 X-ray photons interact with atoms.

The attenuation coefficient against $\lg(\text{photon energy})$ graphs for simple scattering (S), photoelectric effect (PE), Compton effect (C) and pair production (PP) are shown below.



With the help of a calculation, explain the minimum photon energy for pair production.

[3]

8 In the 1800s, the atom was considered to be a fundamental particle. It was an indivisible particle of matter. Modern physics shows that this idea is not correct.

Describe the fundamental particles within an atom of carbon-14 ($^{14}_6\text{C}$).
In your answer state the composition of the hadrons.

[4]

9 The table below shows the quark compositions of four particles A, B, C and D.

A	B	C	D
u u d	u d d	u d s	s s s

Which particle has a positive charge?

Your answer

[1]

10 The chemical composition of ancient rocks found on the Earth can be used to estimate the age of the Earth.

Nuclei of rubidium-87 ($^{87}_{37}\text{Rb}$) decay spontaneously into nuclei of strontium-87 ($^{87}_{38}\text{Sr}$).

The half-life of rubidium-87 is 49 billion years.

i. Name the two leptons emitted in the decay of a rubidium-87 nucleus.

1. _____

2. _____

[1]

ii. The percentage of rubidium left in a sample of an ancient rock is 95%.

Estimate the age of the Earth in billion years.

age = billion years [3]

- 11 Fig. 21 shows stable and unstable nuclei of some light elements plotted on a grid. This grid has number of neutrons N on the vertical axis and number of protons Z on the horizontal axis.

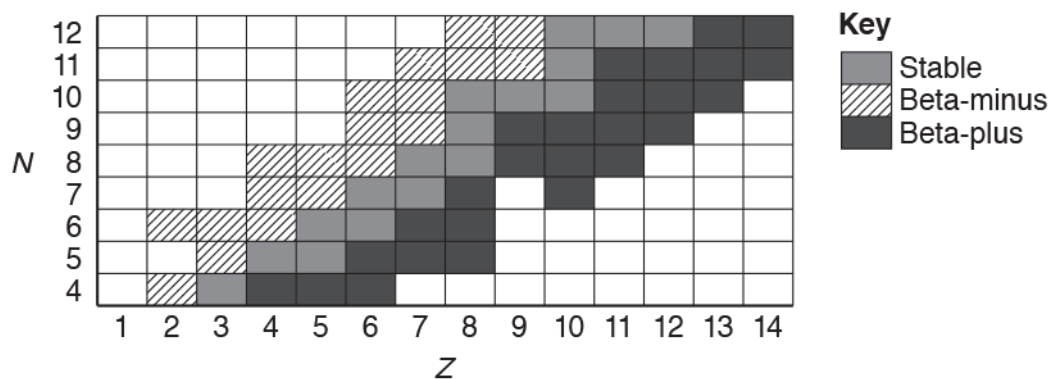


Fig. 21

The key on Fig. 21 shows whether a nucleus is stable, emits a beta-plus particle or emits a beta-minus particle to become stable.

For $Z = 7$, suggest in terms of N why an isotope may emit

- i. a beta-minus particle

----- [1]

- ii. a beta-plus particle.

----- [1]

- 12 Which lepton is emitted in the decay of an up quark and is affected by a magnetic field?

- A neutrino
B electron
C positron
D antineutrino

Your answer

[1]

13(a)

Describe the nature of the *strong nuclear force*.

----- [2]

(b)

- i. Name a hadron found in the nucleus of an atom and state its quark combination.

name of hadron: _____ quark combination: _____ [1]

- ii. Write a decay equation in terms of a quark model for beta-minus decay.

[2]

- (c) The radius of a nucleus is directly proportional to $A^{1/3}$, where A is the nucleon number.

The mass of a proton and a neutron are similar.

Explain why the mean density of all nuclei is about the same.

----- [2]

14 Which statement is correct?

- A Hadrons are made up of protons and neutrons.
- B A positron and a proton are examples of leptons.
- C The positron and the electron have the same mass.
- D The weak nuclear force is responsible for alpha-decay.

Your answer

[1]

15 Write a decay equation for beta-minus in terms of a quark model.

[2]

16 A positive pion (π^+) is an unstable particle produced when high-speed hadrons collide in particle accelerators. The π^+ particle has a charge of $+e$.

What is the quark combination of the π^+ particle?

A $u \bar{u}$

B $u \bar{d}$

C $d \bar{u}$

D $d \bar{d}$

Your answer

[1]

17 Which is the correct decay of a quark?

A $u \rightarrow d + {}^0_{-1}e + \bar{\nu}_e$

B $u \rightarrow d + {}^0_{-1}e + \nu_e$

C $d \rightarrow u + {}^0_{-1}e + \nu_e$

D $d \rightarrow u + {}^0_{-1}e + \bar{\nu}_e$

Your answer

[1]

18 Two leptons are emitted when a down quark decays into an up quark.

Which of the following is correct about this decay?

	force responsible for the decay	leptons emitted
A	strong nuclear	positron and antineutrino
B	weak nuclear	positron and neutrino
C	strong nuclear	electron and neutrino
D	weak nuclear	electron and antineutrino

Your answer

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a	i	0 / neutral	B1	Unit not necessary, but penalise incorrect unit <u>Examiner's Comments</u> Most candidates were able to identify the charge. Most who gave incorrect responses answered a charge of -1, presumably due to the β^+.
		ii	(electron) neutrino	B1	Do not allow antineutrino Ignore symbols If answer line is blank, then the name in the missing space in the equation can be taken as the answer <u>Examiner's Comments</u> Most candidates appreciated that this was a neutrino, with some giving the symbol in the equation too. There were relatively few antineutrino responses with electron being the most common incorrect answer. There seemed to be little connection between this and the previous question for several candidates who correctly gave the charge in (d)(i) but then named a charged particle here, or vice versa.
		iii	4	B1	<u>Examiner's Comments</u> Most candidates were able to determine the correct number of quarks; the most common incorrect response was six, being the total number of quarks.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b		$d \rightarrow u$ $+ {}^0_{-1}e^{-} + \bar{\nu}$	B1 B1	Correct change of quark Allow udd to uud Correct symbols for other two particles Allow e^{-} or ${}^0_{-1}e$ for electron Allow beta symbol for electron Numbers for the neutrino do not need to be present, but must be correct if given ${}^0_0\bar{\nu}$ Examiner's Comments Candidates needed to show the quark change for a single mark, being a down quark to an up quark. This could be done in a number of ways, as long as there was one fewer d and one more u on the right hand side of the equation. This was done fairly well but the other particles were less clear. A neutrino was often given instead of an antineutrino and even though a beta-minus was given in the question, this was often written as a beta plus. A significant number of candidates gave no response to this question. This did not seem to be an issue of time.
			Total	5	
2			D	1	Examiner's Comments This relatively simple question was correctly answered by of the vast majority of candidates. The most common distractor was B which was presumably due to a lack of certainty of quark charges.
			Total	1	
3			D	1	Examiner's Comments Most candidates were able to correctly identify the classification and force. To reduce their options, many candidates circled the correct force.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4			C	1	<u>Examiner's Comments</u> This question was correctly answered by most of the candidates and demonstrated a good knowledge of ionising radiations in medical applications.
			Total	1	
5			C	1	<u>Examiner's Comments</u> A good proportion of candidates were able to answer this correctly. Good practice in this style of questions is to tick the statements that the candidates are sure is correct and try a 'best fit'. Most candidates knew statement 3 was correct and so response B was fairly uncommon as an incorrect response.
			Total	1	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
6	a	Mark first two named forces only Strong (nuclear force) Any one from: short-range / range of about 3 fm / attractive between 0.5 fm to about 3 fm / repulsive below 0.5 fm Gravitational (force) Any one from: long (infinite) range / attractive / negligible / follows inverse square law	M1 A1 M1 A1	Ignore any comments on the weak nuclear force Penalise any clearly incorrect statements. Implication that strong is always attractive will lose A mark Allow 'gravity' Penalise any clearly incorrect statements <u>Examiner's Comments</u> There were many superb answers to this question, with candidates giving plenty of correct detail in describing these forces. The vast majority appreciated the strong nuclear force was one of the forces and were able to give correct numerical values in terms of attractive and repulsive distances. Some stated that it was a purely attractive force, up to about 3fm, which could not be given the nature/range mark. Those who selected gravitational force as their second force were generally able to score the nature/range mark as this was simpler than that required for the strong force. A number of candidates stated that the electromagnetic force acted on the two hadrons and tried to justify the charge of the neutron in terms of its quarks.

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
b	Level 3 (5–6 marks) Clear description including at least one correct comment on quarks and analysis of density <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Some description and some analysis of density or clear description or clear analysis of density <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description or Limited analysis <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 marks <i>No response or no response worthy of credit.</i>	B1 × 6	Indicative scientific points may include: Description (max L2 if either alpha or beta not described) ${}^{14}_6\text{C} \rightarrow {}^0_{-1}\text{e} + {}^{14}_7\text{X} \text{ (ignore identification of X)}$ • Before: 6 protons and 8 neutrons • After: 7 protons and 7 neutrons • Neutron → u d d and/or Proton → u u d • Neutron changes into proton (+ electron + antineutrino) • Down quark changes into an up quark (+ electron + antineutrino) ${}^{235}_{92}\text{U} \rightarrow {}^4_2\alpha + {}^{231}_{90}\text{Y} \text{ (ignore identification of Y)}$ • Before: 92 protons and 143 neutrons • After: 90 protons and 141 neutrons • 2 neutrons and 2 protons removed (by alpha particle) • Has 6 fewer u quarks and 6 fewer d quarks Density (max L2 if analysis done without use of data) • masses: 14.0(u) and 235.0(u) • ratios: masses = 16.8; radii = 2.55; volumes = 16.6 • volume ~ 2.9³ and 7.4³ (ignore 4/3π and 10⁻¹⁵) • densities calculated (about 2.3 × 10¹⁷ kgm⁻³) • density shown to be (roughly the) same Examiner's Comments This is the second level of response (LoR) question in this paper, and was designed to assess nuclear and particle physics, specifically sections 6.4.1 and 6.4.2 of the specification. In particular, there are two parts (description and analysis) relating to two radioisotopes, and some given data. As previous, there are no specific marking points and examiners use a “best-fit” approach to the marking. Indicative scientific points are given in the mark scheme as a guidance what to expect, but not all need to be satisfied to be given full marks. Incorrect physics can be expected to be penalised however. Candidates should always be encouraged to structure their answers carefully and in calculations

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			aim to give explanation of each stage rather than just producing a number. The two separate sections can be separated into a description of the decays, and an analysis of the data to show the two nuclear densities are the same. The key points in the description that examiners were looking for were:—numbers of protons before and after each decay – a description of the change in quark structure during beta decay – correctly balanced equations for alpha and beta-minus decay – the nature of the emissions in each decay. The key points in the analysis that examiners were looking for were: – calculation of volumes of nuclei, using $(4/3\pi) r^3$ – correct use of the density formula – numerical evaluation of the densities of the nuclei and a comparison. The descriptions of the decays varied hugely both in quality and quantity. Most candidates were aware that a beta particle is an electron, but several seemed to think it was an orbiting one which had no structural changes on the nucleus. Those who did know it was from a neutron to proton change often mixed up the nucleon numbers afterwards. Alpha decay was generally done better with most candidates knowing the nature of an alpha particle and correctly stating the structure of the remaining nucleus. Attempts were generally made to changes in quark structures (mostly for beta decay) and the structure of a neutron and proton were known, although strangely several candidates stated the change was from an up quark to a down quark. Candidates who did not give any descriptions of the quark structures were limited to Level 2. For the most part, the analysis was done better than the description and candidates were generally able to correctly evaluate the two densities and make a suitable comparison. Units of density were not penalised here, with many candidates using the mass (in u) in the calculation but giving the units of density as kg m^{-3}, although many stating u m^{-3}. However, many did correctly calculate the mass of the nucleus using the correct factor

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			coming out with densities of approximately $2.3 \times 10^{17} \text{ kg m}^{-3}$. Those who took an approach to algebraically show that the density of every nucleus is independent of the nucleon number (using a standard method to get to some variation of $3u/4\pi r^3$) alone were unable to reach Level 3 as they had not used the data. The use of r_0 as $1.2 \times 10^{-15} \text{ m}$ was commonly seen. ${}^6_6\text{C} \rightarrow {}^6_7\text{N} + {}^0_{-1}\text{e} + \bar{\nu}_e$ beta $\rightarrow$ neutron neutron $\rightarrow$ proton neutron = UDD & proton = UUD UDD In the beta minus decay, 1 down quark turns to a up quark and one proton turns into a proton A candidate who provides some description and analysis which places it in Level 2, but the confusion over alpha decay and general lack of clarity means it would be at the lower point (3 marks).
	Total	10	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
7			(energy =) $9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$ (energy =) $2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2 / 1.60 \times 10^{-19}$ $\lg 1.0(2) \times 10^6 = 6$ (as on graph) OR (energy =) 1.0×10^6 (eV) or $\lg 1.0 \times 10^6 = 6$ (from graph) (energy =) 1.6×10^{-13} J and evidence of mc^2 $2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2 \approx 1.6 \times 10^{-13}$	B1 B1 B1 B1 B1 B1	Note this is 8.2×10^{-14} (J) Note this is $1.0(2) \times 10^6$ eV Note this can be shown in a variety of ways
			Total	3	
8			Electrons and quarks identified as fundamental particles There are 6 electrons, 6 protons and 8 neutrons Composition of proton $\rightarrow$ u u d Composition of neutron $\rightarrow$ u d d	B1 B1	Allow e for electron, p for proton, and n for neutron throughout Allow 6 electrons, 20 u and 22 d Do not award this mark if electron has quark-composition Allow '2 up and 1 down' Allow '2 down and 1 up'
			Total	4	
9			A	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
10		i	Electron and (electron) antineutrino	B1	Allow beta-minus (particle) / β^- ; $\bar{\nu}_{(e)}$ Allow anti electron neutrino <u>Examiner's Comments</u> The majority of the candidates scored a mark for electron and antineutrino as the two leptons. The most common incorrect answers were <i>neutrino</i>, <i>positron</i>, <i>proton</i> and <i>neutron</i>. The pairing of electron and positron also appeared on some scripts.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	$\lambda = \ln 2 / 49$ or $\lambda = 0.0141$ (billion y^{-1}) $0.95 = e^{-0.0141t}$ or $0.95 = e^{-4.48 \times 10^{-19}t}$ (age = $-\ln(0.95)/0.0141$) age = 3.6 (billion years)	C1 C1 A1	Allow $\frac{\ln 2}{49 \times 10^9 \times 3.16 \times 10^7}$ or $4.48 \times 10^{-19} \text{ (s}^{-1}\text{)}$ Allow both C1 marks for $\ln(0.95) = \ln(0.5) \times t/49$ Allow $0.05/0.0141 \approx t$ (this gives 3.5.. for the final mark) Note age in seconds is $1.15 \times 10^{17} \text{ (s)}$; this will score 2 marks Examiner's Comments Many of the top-half candidates demonstrated how the age of the Earth could be calculated in just a few lines. The use of natural logs ($\ln$) was faultless. Most candidates calculated the decay constant and then used the equation $0.95 = e^{-\lambda t}$, or its equivalent $\ln 0.95 = -\lambda t$, to calculate the age t. Candidates are reminded not to round numbers in long calculations – it is good practice to keep all the digits on your calculator. A significant number of candidates rounded the decay constant to 2 SF ($0.014 \text{ billion } y^{-1}$), and this gave an answer of 3.7 billion years. The correct answer, without rounding $\ln 2/49$, was 3.6 billion years. On this occasion, examiners allowed the 3.7 billion years answer. The command term 'estimate' in the question made a small number of candidates to use the equation $\frac{\Delta N}{\Delta t} \approx -\lambda t$. This was allowed, and it gave an estimated age of 3.5 billion years.  Misconception These were some common errors being made in this question, these are summarised below. • Incorrect conversion of billion years into seconds. (Most candidates calculated the decay constant in billion y^{-1}, which easily led to the correct answer in billion years.) • Using $\ln 0.05$ instead of $\ln 0.95$ when calculating the age of the Earth.
			Total	4	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
11		i	Too many N / neutrons	B1	Allow 'neutron-rich' or (for stability) neutron changes to proton or (for stability) charge increases / Z changes to 8 Allow too few protons / 'proton-poor' <u>Examiner's Comments</u> This question required analysis of the information provided in Fig. 21. Most candidates scored a mark for either recognising that the isotope had too many neutrons or a neutron had to decay into a proton in order to provide stability.
		ii	Too few N / neutrons	B1	Allow 'neutron-poor' or (for stability) proton changes to neutron or (for stability) charge decreases / Z changes to 6 Allow too many protons / 'proton-rich' <u>Examiner's Comments</u> A range of answers were allowed in this question requiring analysis of Fig. 21. Most candidates scored a mark for either recognising that the isotope had too few neutrons or a proton had to decay into a neutron in order to provide stability.
			Total	2	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
12			C	1	
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
13	a		Any <u>two</u> from: It acts between quarks / nucleons / hadrons 'Short-range' force Repulsive below (about) 0.5 fm Attractive up to (about) 3 fm	B1×2	Allow any correctly named particle Allow any value between 0.5 fm and 5 fm Examiner's Comment Most candidates scored two marks and knew a great deal about the strong nuclear force.
	b	i	proton = u u d or neutron = u d d	B1	Examiner's Comment The modal score here was one mark. The answers were brief with either proton as uud or the neutron as udd. The up ↑ and down ↓ arrows were allowed as acceptable notation for the up and down quarks respectively.
		ii	$d \rightarrow u + {}^0_{-1}e$ $+ \bar{\nu}_{(e)}$	M1 A1	Allow the equation expressed in words Allow $udd \rightarrow uud + {}^0_{-1}e$ Allow ${}^0_{-1}\beta$ Not e^- for electron Allow this mark if electron written as e^- or p^- Examiner's Comment A variety of answers for the decay equations were accepted with most candidates picking up marks. No credit could be given for showing the decay of a neutron into a proton because of the absence of the quarks. Some of missed opportunities were: - Representing the electron as e^- rather than ${}^0_{-1}e$. - Confusing the positron and the electron. - Assuming the decay was $u \rightarrow d$ rather than $d \rightarrow u$.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		mass (of nucleus) $\propto A$ volume (of nucleus) $\propto \text{radius}^3 \propto A$ and clears steps using $\rho = m / V$ to show density is (about) the same	B1 B1	Allow mass = Am, mass $=Au$, etc. Allow r or R for radius Allow any sensible constant in front of the r^3 Examiner's Comment This proved challenging for most candidates with answers lacking clarity. Some candidates secured a mark for suggesting the $\text{mass}_{\text{nucleus}} \propto A$. Only the very top-end candidates managed to show how the density equation and volume $\propto A$ led to the expected conclusion. Too many scripts had vague answers such as 'neutrons and protons are the same, so their density is the same' and 'protons and neutrons have negligible mass so density is unaffected'.
			Total	7	
14			C	1	
			Total	1	
15			$d \rightarrow u + {}^0_{-1}e$ $+ \bar{\nu}_{(e)}$	B1 B1	Allow ${}^0_{-1}\beta^{(-)}$ for the electron
			Total	2	
16			B	1	
			Total	1	
17			D	1	
			Total	1	
18			D	1	
			Total	1	